Sahaj Bhadja

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Summary

Software engineer with 2+ years building **Python/C++ systems**, **PyTorch and TensorFlow pipelines**, computer-vision applications, and scalable ML tooling; autonomous-driving research improved trajectory planning 18% across 100K+ records.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*

- Built **Python data and orchestration services** for an AI platform supporting 500+ users, optimizing indexed SQL access patterns to reduce dashboard latency from 1.2 seconds to 350 milliseconds.
- Restructured **fault-tolerant asynchronous LLM workers** using FastAPI, processing 1,000+ jobs with explicit state transitions and failure isolation while tripling throughput under concurrent workloads without cascading task failures.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*

- Engineered a **computer-vision grading pipeline** using Python, multimodal models, and OCR to evaluate 10,000+ text and handwritten submissions daily with 90%+ consistency across production workloads for instructors.
- Designed **concurrent batch processing** with multithreading and secure PDF, DOCX, and ZIP ingestion, processing 1,000+ files per batch through fault-tolerant model execution and REST APIs in production.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*

- Built a **GPT-based Decision Transformer** for offline RL in autonomous driving, improving long-horizon trajectory planning by 18% in simulation and developing end-to-end pipelines processing 100K+ trajectory logs.
- Built an **HD-map encoder** combining road geometry and ego trajectories through attention, improving closed-loop planning stability 15% across rigorous autonomous-driving simulations and model evaluations for robust behavior analysis.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Engineered **cross-platform data-sharing software** with microservices and automated VM testing across Windows, Linux, and macOS, maintaining reliable nightly builds through Jenkins, PyTest, Mockito, and 20+ unit/integration suites.
- Optimized **large-scale APIs and data pipelines** using OOP, AWS Glue, S3, SQL, indexing, and caching, improving response times 30% and ETL efficiency 68% across production workloads at scale.

Skills

Languages: **Python**, **C++**, C, Java, JavaScript, TypeScript, SQL, Bash
ML & Vision: **PyTorch**, **TensorFlow**, Transformers, Computer Vision, Deep Learning, CNN, RNN, LSTM, Feature Engineering
Autonomy: Decision Transformers, Offline RL, Multi-head Attention, Spatial-temporal Modeling, HD Maps, Simulation
Data & APIs: PostgreSQL, MongoDB, FAISS, REST APIs, FastAPI, Flask, data pipelines, ETL, vector search
Platform: Docker, Linux, AWS, Azure, Git, GitHub Actions, CI/CD, PyTest, Jenkins

Projects

CaptionCraft Live | *PyTorch, Transformers, Flask, Computer Vision*

- Benchmarked **BLIP, GIT, and ViT-GPT-2 vision-language models** on 10K images and 100K captions, improving caption accuracy from 81% to 96% while building APIs for scalable real-time inference.

PiXMod Live | *PyTorch, CLIP, SAM, Stable Diffusion*

- Built an **AI image-processing platform** using SAM for segmentation, Stable Diffusion for generative editing, and CLIP for semantic retrieval across 100K+ indexed image embeddings and 1K+ edited assets.

CPU Scheduling Simulator Live | *C++, React, Algorithms, Visualization*

- Implemented **six CPU scheduling algorithms** in C++ with React controls and Gantt visualization, validating 1K+ process combinations while reducing average turnaround time 25% and improving debugging efficiency 20% across simulated workloads.